Essential Maths for Physics Undergrads

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About the book

Essential Maths for Physics Undergrads aims to introduce basic, and crucial mathematical areas and tools required for any physicist. No technical knowledge is assumed past the point of high-school science and mathematics. This book may serve as a pre-university text for those who will later go on to study a scientific discipline (not necesserily just maths and physics, although some areas may be most useful to students of these subjects) or as a semi-technical source for curious readers.

The layout of the book will be similar to what you might find in a university-level mathematics module. Concepts will be introduced with clear definitions via **Definition**, **Theorem**, **Corollary**... etc... markers. This may also help students who are not yet aquainted with such conventions to get used to them before they begin their university studies.

Each chapter will include many differnt examples along with the above mentioned flags. Some practice questions will also be included at the end of each chapter. These will, however, be limited.

Chapters in this book may be used separately from each other as there will be minimal overlap in material. It should be said, however, that titles of all chapters should be mentally prefixed with 'Introductory' as no advanced concepts are covered.

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Introduction

This present chapter covers the infamous area of mathematics which arguably forms the very bedrock of modern physics. In fact, it indeed forms the basics of all modern hard sciences. Originally developed (or discovered) by Newton and Leibniz (full credit may be awarded to either party depending on your preference), Calculus is concerned, generally, with the study of continuous functions. In this chapter, we will first look at the idea of a limit (a way of evaluating functions at 'undefined' points). Then, we will use this new concept to define and use derivatives and integrals. On the way, we will also touch on the notion of operators which will be important in later chapters as well.

3.1 Limits

3.1.1 Defining limits

We begin by defining a limit. You may have encountered functions in high-school maths which seemed to be undefined for certain values. If you have, there is a good chance that your teacher(s) simply told you to not worry about these values, or that these values are anomolous, or something along those lines. However, for some functions, you can define them at these 'anomolous' points. How? We employ limits...

As an example, consider the function below

$$f(x) = \frac{x^2 - 4}{x - 2} \tag{3.1}$$

Now, substitute in $x = 2$. You will notice that in this case, we get "0/0" which you would rightly say is nonsense (or undefined). However, you may be surprised to learn that this function is actually perfectly defined at $x = 2$. To see this, start substituting in values of x into $f(x)$ which are not 2, but rather are very close to 2. For example, substitute in the numbers 1.8, 1.9, 1.95, 1.99... and so on. You will notice that the answer you get starts to get closer and closer to (but never quite reaching) 4. This is in fact the basic concept of a limit. A bit more rigorously, we may define a limit in the following way.

Definition: *(Limit)* Let f be a function of a real variable $x \in \mathbb{R}$ and let $u \in \mathbb{R}$ be a point at which f is undefined. Then, the limit of f as x approaches u is

$$\lim_{x \to u} f(x) = L \tag{3.2}$$

where $L \in \mathbb{R}$.

Note: the above definition is a special case and in fact, limits can be defined for different vector spaces, not only $\mathbb{R}$.

Example: Consider the function

$$f(x) = \frac{1}{x} \tag{3.3}$$

Let's think about the value of $f(x)$ at infinitely large x. You will be quick to notice that it's once again undefined as $1/\infty$ is mathematical giberish. However, now let's consider

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{1}{x} \tag{3.4}$$

the limit as x approaches ∞. For larger an larger x, $f(x)$ becomes smaller. In fact, for an arbitrarily large x value, the value of our function apporaches 0. Therefore, we can say with confidence that

$$\lim_{x\to\infty} \frac{1}{x} = 0 \tag{3.5}$$

Notice the subtle but crucial difference between saying that $1/\infty = 0$ and the eq.(2.5). We are not saying that division by ∞ is 0, because again, this doesn't make much sense for multiple reasons. One of them being that ∞ is not a number, so you cannot treat it as such. What we *are* saying, however, is that for larger and larger values of x this funcion $f(x) = 1/x$ *approaches* 0.

3.1.2 Undefined limits

Limits have proven quite a useful tool for evaluating functions at 'undefined' points. However, there are some functions which are not defined at particular points. That is, it's not always guaranteed that any limit of any function will be defined. These are called undefined limits or undefinable limits.

Example: Consider the function

$$f(x) = \sin x \tag{3.6}$$

How would we go about finding the value of f for an infinitely large value of x? Well, we would employ the usual strategy of trying to find the limit of f as x approaches infinity. However, sin is a periodic function. This means it never settles on a value as it extends to positive and negative infinity. It will always oscillate between $+1$ and -1. Therefore, we say that the limit

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \sin x \tag{3.7}$$

is undefined. Or that the limit doesn't exist.

3.1.3 One-sided limits

3.2 What is an Operator?

3.2.1 General operators

To make the next sections a little more stright forward, we will spend this section defining operators. There are, broadly speaking, two kinds of operators: linear and non-linear. First, however, let's define a general operator.

Definition: *(Operator)* An operator, $\hat{O}$, is a mapping of a space S, called the domain of $\hat{O}$, onto another space S^*, called the co-domain of $\hat{O}$, s.t. S and S^* may or may not be the same space.

The above definition is the closest possible general definition of operators. Mathematicians don't usually write such a broad definition of operators on account of how many different ways exist of interpreting such objects exist. For example, if we consider the space $\mathbb{R}$, then any function, f, of real variables is considered an operator with $S = S^* = \mathbb{R}$.

Example: Let $M \in \mathbb{R}^{n\times n}$ and let $S = \mathbb{R}^n$. Then, M is an operator on S with co-domain $S^* = S$.

We will mostly encounter operators which map spaces onto themselves. That is to say their domain and co-domain are the same space. However, the domain and co-domain are allowed to be different spaces. An example of this would be a matrix with complex entries acting on real vectors (example below).

Example: Let $M \in \mathbb{C}^{n\times n}$ and let $S = \mathbb{R}^n$. Then, M is an operator on S with co-domain $S^* = \mathbb{C}^n$.

The concept of operators is of major importance in calculus as we will be dealing with special operators (namely derivatives and integrals) which act over the space of functions. In fact, these two operators will turn out to be inverses of each other, which is, in essence, the statement of the fundamental theorem of Calculus. Therefore, we need to address the concept of inverse operators before we move on to the concept of linearity.

Definition: *(Inverse Operators)* Let $\hat{O}$ be an operator with domain S and let $f \in S$ be a member of S. Then, there exists an operator, $\hat{O}^{-1}$ called the inverse of $\hat{O}$ s.t.

$$\hat{O}\hat{O}^{-1}f = \hat{O}^{-1}\hat{O}f = f \tag{3.8}$$

for all $f \in S$.

Example: Consider the operator

$$P = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix} \tag{3.9}$$

acting over $\mathbb{R}^2$. Then, the inverse of P, P^{-1} is the operator

$$P^{-1} = \frac{1}{6}\begin{pmatrix} 3 & 0 \\ -1 & 2 \end{pmatrix} \tag{3.10}$$

3.2.2 Linear Operators

There is a special class of operators called linear operators. Concepts associated with this class are especially important for our puposes in this chapter as we will later see (and prove) that both derivatives and integrals are examples of linear operators.

Definition: *(Linear operator)* Let $\hat{O}$ be an operator with domain S. Let also a, b be constants and $f, g \in S$ be members of the domain of $\hat{O}$. $\hat{O}$ is called a linear operator iff (if and only if)

$$\hat{O}(af + bg) = a\hat{O}f + b\hat{O}g \tag{3.11}$$

for all constants a, b and $f, g \in S$.

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3.4 Integrals

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